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Tailored Application Package: AI Research &
Systems Intern

Applicant: Alex Rivera

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Target Role: AI Research & Systems Intern (AnthroMetrics AI Research)

Date: September 2026

1. Executive Skill Gap & Alignment Analysis

Competency Area

AnthroMetrics JD Requirement

Alex Rivera Experience & Match

Alignment Level

Local LLM & Inference

Experience with quantized SLMs,

vLLM, Ollama, ONNX

Engineered sub-80ms semantic RAG

engine at VeriLocal Systems;

benchmarked 7B SLMs at Cognition Labs

EXACT MATCH

(100%)

Agent Safety &

Verification

Sandboxed tool-use, safety rails,

verifiable execution

Created SecureAgent runtime; built

human permission gates & filesystem

post-action verifiers

EXACT MATCH

(100%)

Retrieval & RAG

Hybrid sparse-dense retrieval,

ChromaDB / FAISS

Implemented hybrid RRF search in

Project Alpha; indexed 50k+ internal

documents

EXACT MATCH

(100%)

Systems & Systems

Code

Python, PyTorch, C++/Rust,

AsyncIO

Advanced Python, intermediate Rust,

B.S. CS from Stanford University (GPA:

3.92)

HIGH MATCH

(95%)

Key Strategic Highlights

- Direct Systems Background: Built deterministic sandboxing frameworks reducing unauthorized file access to 0%.
- Relevant Stanford Coursework: CS224N (Natural Language Processing), CS231N (Computer Vision), CS229 (Machine Learning), CS110 (Computer Systems).
- Quantized Benchmark Expertise: Prior experience evaluating tool precision and JSON adherence in 7B/8B local models.

2. Customized Cover Letter

To: Hiring Team, AnthroMetrics AI Research

Subject: Application for AI Research & Systems Intern — Alex Rivera

Dear Hiring Team at AnthroMetrics AI,

I am writing to express my enthusiastic interest in the AI Research & Systems Intern position. Having tracked AnthroMetrics' pioneering contributions to verifiable intelligence and reliable agent architectures, I believe my background in local inference optimization, deterministic tool sandboxing, and hybrid retrieval systems aligns directly with your

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mission.

During my time as an AI Systems Engineer at VeriLocal Systems, I spearheaded the development of an on-device RAG engine indexing over 50,000 documents with sub-80ms latency. More importantly, I designed and implemented human-in-the-loop permission gateways and post-action disk verifiers, guaranteeing that autonomous actions remain fully deterministic, inspectable, and secure.

Prior to that, during my internship at Cognition Labs, I developed quantitative evaluation frameworks for local quantized models (Qwen, Llama, Mistral), achieving 94.2% valid tool-dispatch syntax. These practical engineering experiences, coupled with my Stanford Computer Science foundation (GPA: 3.92), have given me a rigorous understanding of both the

theoretical and systems-level challenges inherent to trustworthy agent design.

I would welcome the opportunity to contribute to AnthroMetrics' state-of-the-art research and systems initiatives. Thank you for your time and consideration.

Sincerely,

Alex Rivera

alex.rivera@cs.stanford.edu | github.com/arivera-ai

3. 4-Week Project & Onboarding Plan

- Week 1: Architecture Onboarding & Benchmark Baselines
 - Profile existing model inference pipelines using Ollama/vLLM across target hardware.
 - Establish automated test harnesses for structured tool call dispatch.
- Week 2: Verification Engine & Sandbox Integration
 - Implement cryptographic post-action verification hooks for workspace mutations.
 - Deploy interactive human approval gates with diff-preview capabilities.
- Week 3: Hybrid Retrieval & RAG Optimization
 - Integrate Reciprocal Rank Fusion (RRF) sparse-dense retrieval over internal documentation.
 - Benchmark retrieval precision and memory footprint against baseline FAISS/ChromaDB indices.
- Week 4: Synthesis & Production Verification
 - Perform stress tests on concurrent agent execution and audit log immutability.
 - Package deliverables with full unit test coverage and reproducible documentation.

Generated and verified on-device by LocalGPT Deterministic Workspace Agent.